Sourav Yadav

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github.com/sourav-ydv | Portfolio

Summary

Final-year CSE student at IIIT Manipur (graduating 2027) who builds and tests LLM agents: tool-calling agents with human approval, RAG chat, and a fault-injection tool that checks agents stay safe when tools fail. WABI 2026 co-author.

Education

Indian Institute of Information Technology (IIIT) Senapati, Manipur — Aug 2023 – 2027 (Expected)

- B.Tech in Computer Science and Engineering | CPI: 8.85 (6th Sem)

Experience

Research Intern, National Institute of Technology Kurukshetra — Nov 2025 – Apr 2026

Built a Python data pipeline to clean, transform, and validate a large, messy real-world dataset (18,104+ genomes) and trained a coverage-aware deep learning model in **PyTorch**.
Evaluated and iterated on the model (**F1-macro = 0.79**, **AUROC = 0.84**); co-authored the resulting paper, accepted at **WABI 2026**.
Tools: Python, PyTorch, NumPy, Git

Publications

- Co-authored "*Biologically Grounded Explainable Antimicrobial Resistance Prediction Using Knowledge Graphs*", accepted at **WABI 2026** (26th Int. Conference on Algorithms for Bioinformatics).
- Submitted: "*NeSy-AMR: Neuro-Symbolic Coverage Routing for Explainable Antimicrobial Resistance Prediction*", under review at **IEEE TCBB**.

Projects

Havoc: Fault Injection and Evaluation for MCP Agents (Python, asyncio) — github-repo | Report

Built a proxy between an agent and its **MCP** tool servers that injects faults (timeouts, truncated or wrong fields, prompt injections) and checks the final database state with 8 safety checks, instead of trusting the agent's own answer.
Added delta-debugging to shrink failing fault sets, re-runs to separate real from flaky failures, and offline CI replays. Over 411 runs on Gemini 3.5 Flash-Lite, found false refund confirmations in 25–48% of runs under tool failures; a one-line prompt change cut this to 0–10% ($p = 0.0004$).
Tools: Python, asyncio, MCP, SQLite, pytest, GitHub Actions

Trip AI Agent (LangGraph, FastAPI, PostgreSQL) — github-repo | Demo-link

Built a tool-using LLM agent (LangGraph ReAct, Groq) with 7 tools over 4 external APIs, PostgreSQL-backed conversation memory, and an `interrupt()`-based human approval step before any booking; React frontend.
Added connection pooling with retries, rate-limit handling, token-budgeted history trimming, and failure-driven prompt rules.
Tools: Python, LangGraph, FastAPI, PostgreSQL, React, REST APIs

Webpage Chat Assistant (RAG, Structured Extraction) — github-repo

Built a Chrome extension with a FastAPI backend that chats with any webpage using RAG (LangChain, Cohere embeddings, vector store), keeps separate memory per tab, and returns typed JSON via Pydantic.
Fixed a production out-of-memory crash with session eviction and hosted embeddings; added a context-budget regression test.
Tools: Python, FastAPI, Pydantic, LangChain, Groq API, Cohere

Technical Skills

LLMs & Agents: LangGraph, LangChain, Tool-Calling Agents, RAG, MCP, Structured Output, Vector Search, Human-in-the-Loop, Prompt Engineering, Groq / Gemini APIs
Evaluation & Testing: Agent Fault Injection, Delta Debugging, Confidence Intervals, pytest, GitHub Actions (CI)
Programming: Python (asyncio, FastAPI, Pydantic, SQLAlchemy, Pandas, NumPy), SQL, JavaScript, C++, C
Backend & ML: REST APIs, JWT, PostgreSQL, SQLite, React, PyTorch, Scikit-learn, XGBoost, SHAP
Tools: Git, GitHub, Docker, Render, Vercel, Jupyter Notebook

Certifications & Languages

Introduction to Information Retrieval (NPTEL, IISER Kolkata): Elite + Gold, 92%, Top 2% — Jan – Apr 2026
Business Intelligence & Analytics (NPTEL, IIT Madras): Elite + Gold, 95%, Top 5% — Jan – Apr 2026
Getting Started with Competitive Programming (NPTEL, IIT Gandhinagar): Elite + Gold, 90% — Jul – Oct 2025
Korean Language: TOPIK Level 1 — Oct 2025